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Program Evaluation and Review Technique (PERT) in Construction Risk Analysis

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Abstract. Construction project scheduling is one of the most critical factors for project success measurement. Critical Path Method (CPM) is commonly used in the construction industry for project scheduling. CPM is a deterministic method that ignores uncertainties in activities through the network， which determines the duration of the projects. Program Evaluation and Review Technique (PERT) is a probabilistic method based on the assumption that the duration of a single activity can be described by a probability density function. PERT takes into account the uncertainties during the construction process to plan, schedule and control complex projects with many uncertainties. The work reported in this paper describes the PERT work process to carry out the construction schedule risk assessment and illustrates the model through case studies. The utilization in the estimate the construction liquidated damage with the uncertainties is performed, which also can be used in the insurance company to calculate the insurance premium.

Introduction

In construction project contracts, it is necessary to define the duties, roles and responsibilities of the each participant in the agreement with three factors that must be identified and clearly defined for procurement success: scope, price and time [1-2]. The owners, general contractors, project managers and even subcontractors are now increasingly realizing the importance of completing the projects satisfactorily on time and maintaining the profitability by implementing new technology and innovative construction process philosophy. Not only does the construction schedule support to plan and manage the project to reach the completion deadline, it is related to legal, economical issues and the performance claims as well [1-2]. Therefore, the construction schedule is a tool for the project participants to coordinate effectively and efficiently to identify the timeline and provide evidences of construction delays and claims. Computer scheduling programs are available for construction scheduling to plan and control the project process. In particular, the Critical Path Method (CPM)) has been rightfully provided a desirable template for the project manager to build an integrated scheduling system. CPM is the most common technique adopted by the construction industry for construction project scheduling. It was developed in 1957 in the context of addressing the challenge of shutting down chemical plants for maintenance and then restarting the plants once the maintenance had been completed [3]. And it was rapidly extended to the application in construction [3]. During the scheduling programs, a few methods are available to measure the work in progress of an activity. Some depend on empirical judgment; others may be calculated

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analytically. Several often used ones include: Schedule Variance, which is to compare budgeted cost of work scheduled and budgeted cost of work performed. Another is Schedule Performance Index (SPI) to indicate that the job is behind the schedule by SPI<1 and the job is earlier than the planned job at SPI>1 [3].

However, all these methods mentioned above are categorized to CPM. CPM assumes that at least one path through the network determines the duration of the whole project and this path is the critical path, the activities in this path are called critical activities. CPM is a deterministic method which does not consider the uncertainty in the project and the completion probability in each activity.

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PERT is a probabilistic method which is based on the assumption that one path determines the duration of the project, but the duration of each activity is characterized by a probability density function. The project duration can be calculated by the sum of the "expected" durations of each activity in critical path. Then, According to the central limit theorem, the project duration can be shown to follow a normal distribution, uniquely defined by the parameters computed from each activity parameters [3-4]. PERT was developed for the U.S. Navy's Polaris project which is a complex project with thousands of contractors. The concept of PERT is based on the assumption that the duration of each activity is a random variable that behaves according to some known and identical probability distribution. The distribution of choice is a beta distribution [5-6]. PERT approach is stated in a few books and papers, but to the best knowledge of the author, report of the application of PERT approach in construction risk assessment is scarce in the literature. Details of the PERT work process are described as follows.

The duration of each activity in PERT is calculated by:

$$t_e = \frac{t_0+4t_m+t_p}{6}. \tag{1}$$

where t_e is the expected time; t_o is the optimistic time; t_m is the most likely probable activity time, and t_p is the pessimistic time. The standard deviation is given by:

$$s_1 = \frac{t_p-t_0}{6} \ . \tag{2}$$

According to the central limit theorem, the sum of a large number of independent random variables follows a normal distribution. Thus, in a construction project which consists of lists of activities, the project duration T will tend to follow a normal distribution with a mean and variance, denoted by μ and σ^2, respectively, given by

$$\mu = \sum_{j=1}^{n} t_j. \tag{3}$$

$$\sigma^2 = \sum_{j=1}^{n} \sigma_j^2. \tag{4}$$

where t_j and σ_j denote the duration and standard deviation of the *j*th activity among the critical ones, respectively.
And the project duration is normal distribution with a probability density function (PDF)

$$f(T) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{1}{2}\left(\frac{T-\mu}{\sigma}\right)^2\right]. \quad (5)$$

This allows for risk assessment analysis of the network. One assumes that the expected length of the whole project is the sum of the critical activities' separate expected lengths. Thus by summing up all the t_es of all critical activities, one can get the duration of the project. Similarly, the variance of the project is equal to the sum of the variances associated with each of all critical activities'. By calculating the mean and the variance, risk analysis can be performed.

Case Study

The procurement of structural steel for a building that will require about 5,000,000 pounds of structural steel elements is given as in Table 1. Each activity in the procurement process follows a normal distribution with the following parameters (see Table 1). The activities, with the exception of the delivery from the mill and detail drawings are sequential. A firm is under contract to complete this work with a liquidated-damages clause in the contract such that the contractor must pay $27,622 per day for all days that are overdue of the scheduled completion date. To ensure the liquidated damages are collectible, the owner has required the contractor to seek insurance against the liquidated damages.

Table 1. Mean and standard deviation of the activities for structure steel building

Activities	Mean	Std Dev
Delivery from the mill	45	5
Detail Drawings	65	10
Fabrication	140	20
Delivery to shake-out yard	35	2
Shake-out	40	5
Delivery to site	13	2

Since delivery from the mill and detail drawings are sequential, the critical path from the network is found of delivery from the mill to fabrication to delivery to shake-out yard to shake-out to delivery to site. Then the total duration of the project is:

$$\mu = \sum m_i = 293. \quad (6)$$

$$\sigma^2 = \sum \sigma_i^2 = 533. \quad (7)$$

Then T follows normal distribution with μ=293 and $\sigma = \sqrt{533} = 23$. The estimated duration for which you have a 0.9 probability of assurance of meeting can be calculates as follows:

$$P(Z \leq x) = P\left(z \leq \frac{x-293}{23}\right) = 0.9. \quad (8)$$

$$x = 323\ days. \quad (9)$$

After investigating the site, interviewing the project manager and chief estimator, the reviewing the procurements of major equipment, if the manager estimates the coefficient of variation to be 0.2. Thus $\sigma = 0.20 \times 293 = 58.6$

Assume the insurance company offer a-45 day deductible period in the insurance contract,

$$E(t) = \int_{338}^{\infty}[\$27{,}622(t-(293+45))]\frac{1}{\sqrt{2\pi}(58.6)}\exp\left[-\frac{1}{2}\left(\left(\frac{t-293}{58.6}\right)^2\right)\right]dt = \$460{,}809. \quad (10)$$

Where E(t) is the expected loss to the insurance company.

From the Central Limited Theory, one knows that the duration of the project will follow a normal distribution. Liquidated damages start for the construction on calendar day 293 and for the insurance company on calendar day 293+45=338. Therefore, the function for the liquidated damages (LD) exposure to the insurance company becomes LD=$27,622(t-338).

Conclusions

CPM approach is a primary tool for planning, scheduling and controlling in construction industry. It consists of the logical relationship between the activities involved in a project. However, CPM is deterministic approach that does not consider the uncertainties associated with all activities in a project. Considerable work is required to develop estimates in a probabilistic framework. In such case, the duration is represented by a random variable. The sum of those random variables follows a specific probability distribution. This paper investigates the PERT work process to carry out the construction schedule risk assessment. Even though there are still some complicate parts for the PERT application, for the project manager and scheduler, they should always remember that a carefully scheduling the project is part of contract, it may be used in court later to prove or disprove a delay claim.

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